

Peer Review: Software Engineer Earnings Analysis

Project Reviewed: Confidence Crew (group 11)

Reviewer: Skajen (Group 4)

Date: 24 Aug 2025

1. Executive Summary

This is an **exceptionally strong and important project**. The team has demonstrated maturity in **methodology** and **communication**. The project has high potential for real-world impact.

To maximize the project's success and ensure a flawless presentation, the team should now focus on finalizing consistency and robustness. Addressing the **KEY PRIORITY** below will ensure the analysis is fully **defensible**, the results are **perfectly reproducible**, and the project delivers maximum impact.

2. Project Strengths

The team deserves recognition for the following outstanding aspects:

- **Impactful Topic and Clear Goal:** The focus on wage equity and disparities across demographics is highly relevant and clearly addressed by the central research question.
- **Gold-Standard Communication :** The strategy is business-ready, with a well-defined audience, actionable objectives, and the perfect choice of an **Interactive Tableau Dashboard** as the main artifact.
- **Methodological Rigour:** The use of **ANOVA** and the exploration of **interaction effects** (e.g., gender times education) demonstrate advanced analytical thinking.
- **Comprehensive Documentation:** Every step is documented with a dedicated README. The variables_definition.md is a highly valuable artifact for reproducibility.
- **Mature Collaboration:** The Group Norms & Collaborative Guidelines document shows excellent project management and team structure.

3. Required Actions:

The following is a comprehensive schedule of actions required to finalize the project for presentation.

<i>Priority</i>	<i>Focus Area</i>	<i>Detailed Action Required</i>	<i>Folder(s)</i>
Critical	Occupation Code Inconsistency	Verify and unify the Software Developer OCC Code (1106 vs. 1020). The Domain Study (0_domain_study/) uses 1106 , but the cleaning and analysis documentation (Folders 2, 4) uses 1020 . The team must verify and fix this inconsistency across all documentation and the cleaning script.	0, 2, 4
Critical	Reproducibility	Code Stability: Conduct a final full audit to ensure the entire analytical pipeline runs flawlessly from start to finish.	
Critical	Data Structure	Raw Data Placement: Move the raw data file from the user-specific folder (anna_data_collection/) to a generic 1_datasets/raw/ subfolder.	1
Medium	Data Summary	Missing Documentation: Complete the "Dataset Summary" section in the 1_datasets/README.md with final sample size and initial statistics (e.g., earnings range).	1
Medium	Organization	Root Clutter: Move presentation-related markdown files like presentation_data_modeling_improved.md and presentation_slides_content.md from the root directory into the 6_final_presentation/ folder.	Root
Medium	Organization	Ambiguous File Naming: Files in 0_domain_study/ like convergent_thinking and divergent_thinking should be renamed to be more descriptive (e.g., initial_brainstorm_notes.md).	0